

fields in CNNs. Wu et al. (2017) tackles the problem of deriving structured representation from raw pixel-level inputs. Dai et al. (2018) combines structured visual representation and theorem proving.

Graph neural networks and relational inductive bias. Graph convolution networks (GCNs) (Bruna et al., 2014; Li et al., 2016; Defferrard et al., 2016; Kipf & Welling, 2017) is a family of neural architectures working on graphs. As a representative, Gilmer et al. (2017) proposes a message passing modeling for unifying various graph neural networks and graph convolution networks. GCNs achieved great success in tasks with intrinsic relational structures. However, most of the GCNs operate on pre-defined graphs with only nodes and binary connections. This restricts the expressive power of models in general-purpose reasoning tasks (Li et al., 2016).

In contrast, this work removes such restrictions and introduces a neural architecture to capture lifted rules defined on any set of objects. Quantitative results support the effectiveness of the proposed model in a broad set of tasks ranging from relational reasoning to modeling general algorithms (as decision-making process). Moreover, being fully differentiable, NLMs can be plugged into existing convolutional or recurrent neural architectures for logic reasoning.

Relational decision making. Logic-driven decision making is also related to Relational RL (Van Otterlo, 2009), which models the environment as a collection of objects and their relations. State transition and policies are both defined over objects and their interactions. Examples include OO-MDP (Diuk et al., 2008; Kansky et al., 2017), symbolic models for learning in interactive domains (Pasula et al., 2007), structured task definition by object-oriented instructions (Denil et al., 2017), and structured policy learning (Garnelo et al., 2016). General planning methods solve these tasks via planning based on rules (Hu & De Giacomo, 2011; Srivastava et al., 2011; Jiménez et al., 2019). The goal of our paper is to introduce a neural architecture which learns lifted rules and handle relational data with multiple orders. We leave its application in other RL and planning tasks as future work.

Neural abstraction machines and program induction. Neural Turing Machine (NTM) (Graves et al., 2014; 2016) enables general-purpose neural problem solving such as sorting by introducing an external memory that mimics the execution of Turing Machine. Neural program induction and synthesis (Neelakantan et al., 2016; Reed & De Freitas, 2016; Kaiser & Sutskever, 2016; Parisotto et al., 2017; Devlin et al., 2017; Bunel et al., 2018; Sun et al., 2018) are recently introduced to solve problems by synthesizing computer programs with neural augmentations. Some works tackle the issue of the systematical generalization by introducing extra supervision (Cai et al., 2017). In Chen et al. (2018), more complex programs such as language parsing are studied. However, the neural programming and program induction approaches are usually hard to optimize in an end-to-end manner, and often require strong supervisions (such as ground-truth programs).

5 CONCLUSIONS AND DISCUSSIONS

In this paper, we propose a novel neural-symbolic architecture called Neural Logic Machines (NLMs) which can conduct first-order logic deduction. Our model is fully differentiable, and can be trained in an end-to-end fashion. Empirical evaluations show that our method is able to learn the underlying logical rules from small-scale tasks, and generalize to large-scale tasks.

The promising results open the door for several research directions. First, the maximum depth of the NLMs is a hyperparameter to be specified for individual problems. Future works may investigate how to extend the model, so that it can adaptively select the right depth for the problem at hand. Second, it is interesting to extend NLMs to handle vector inputs with real-valued components. Currently, NLM requires symbolic input that may not be easily available in applications like health care where many inputs (*e.g.*, blood pressure) are real numbers. Third, training NLMs remains nontrivial, and techniques like curriculum learning have to be used. It is important to find an effective yet simpler alternative to optimize NLMs. Last but not least, unlike ILP methods that learn a set of rules in an explainable format, the learned rules of NLMs are implicitly encoded as weights of the neural networks. Extracting human-readable rules from NLMs would be a meaningful future direction.

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